

Virtual Machines, Isolated Labs, Powershell Scripting, and Active Investigation

These notes were written by Kevin Beideman. The recommended time to understand and practice these concepts is 4 hours.

Videos: <https://www.youtube.com/watch?v=-uDYvy2jQzM>

Articles:

Virtual Machines + Home Lab Foundations

What is virtualization?

The process of creating a virtual (software-based) version of something instead of using a physical version directly. With one laptop using virtualization, you can run: Windows, Linux, and another copy of Windows all at the same time. Some key components:

1. *Host Machine: the real physical computer*
2. *Hypervisor: software that creates and manages virtual machines*
3. *Virtual Machine: software-based computer with its own OS, CPU, memory, storage, network settings*

What is a virtual machine? What is a hypervisor?

So basically, a VM is a computer running INSIDE another computer.

What is Type 1 vs Type 2 hypervisor?

A Type 1 hypervisor runs directly on the physical hardware. Hardware -> Hypervisor -> Virtual Machines. A Type 2 hypervisor runs on top of a normal operating system. Hardware -> Host OS -> Hypervisor -> Virtual Machines. Ex: VirtualBox, VMware Workstation

What is the difference between host OS and guest OS?

The host OS is the main operating system installed directly on the physical computer. It controls: real hardware, files, keyboard/mouse, memory, storage, and networking. The guest OS is an operating system running inside a virtual machine. It only exists inside the host system through virtualization.

What is dynamically allocated storage?

Memory that a program requests while the program is running, instead of being fixed beforehand.

What is NAT mode?

Network Address Translation mode is a networking setup where the VM shares the host computer's internet connection through a virtual router. The VM can access the internet, but devices outside usually cannot directly access the VM.

What is Bridged Networking?

VM networking mode where the virtual machine connects directly to the same network as the host computer. The VM acts like a completely separate device on the network. The VM gets its own IP address from the router, so other devices on the network can communicate directly with it.

What is Host-Only networking?

VM networking mode where the virtual machine can communicate with the host computer, but not directly with the internet or the outside local network.

What are VM snapshots?

VM snapshots is a saved state of a virtual machine at a specific moment in time. It allows you to "freeze" the VM and later restore it exactly how it was.

Windows + Linux Environment Investigation

Run: **whoami** in Ubuntu VM. Remember enumeration is the active process of gathering detailed information about a target system, network, or application to identify potential vulnerabilities and attack vectors.

The whoami command in Linux displays the username of the currently logged-in user.

Next: **pwd**. This stands for Print Working Directory. Displays the full path of the current directory you are currently inside.

Run: **ipaddr**. Then: **ps aux**. This command shows a detailed list of currently running processes on a Linux system.

Lastly: **ss -tulnp**. This is used to display network sockets and listening ports.

What is the difference between netstat and ss?

Both are used to view: network connections, open ports, listening services, and socket information. But, ss is the newer, faster replacement for netstat.

Run: **ls /**. This will list the contents of the root directory in Linux. The **ls** command stands for list, which is like **dir** in Windows.

The /etc directory in Linux contains system-wide configuration files. If Linux were a company: /etc is the filing cabinet containing all the important rulebooks and configuration documents for how everything operates.

The /var directory stores variable data; files that frequently change while the system is running. Commonly stored things are: Logs (system logs), Mail (email storage), Databases (app/service data), Cache, Spool files (print queues/jobs), and Web server data

The /bin directory contains essential command binaries. These are important programs needed for the system to function and for basic command-line usage. Examples: ls, cp, mv, cat, pwd, rm, etc....

The /usr directory contains: user programs, applications, libraries, documentation, and additional system utilities.

Now: run **id**. The id command displays information about a user's identity and permissions. It shows the user ID, group ID, groups the user belongs to.

Next: run **sudo -l**. Sudo stands for superuser do. It allows a user to run commands with elevated privileges. The -l means list privileges. So, this lists the commands the current user is allowed to run using sudo.

What is root in Linux?

In Linux, root is the highest-privileged user account; it is the system administrator with complete control over the operating system. So, root has unlimited permissions.

What are Linux permissions?

Rules that control: who can read files, who can modify files, and who can execute programs.

Read = r, Write = w, Execute = x.

Linux assigns permissions to: User (u), Group (g), Others (o)

Now, run **top**. The top command displays a real-time view of running processes and system performance. Similar to Task Manager on Windows.

Powershell Scripting Foundations

What is scripting?

The process of writing small programs called scripts that automate tasks.

Why is automation important in cybersecurity?

Modern systems generate too much data and too many events for humans to handle manually. Analysts would manually inspect everything, attacks would spread too quickly, and repetitive tasks would waste huge amounts of time.

I'm creating an empty script file in Windows. Trying the command: **New-Item process_report.ps1**.

```
Get-Process |  
Sort-Object CPU -Descending |  
Select-Object -First 10 Name, CPU, WorkingSet |  
Export-Csv top_processes.csv -NoTypeInfoInformation
```

```
Write-Host "Process report generated."
```

Get Process | gets a list of all currently running processes on the system.

Sort-Object CPU -Descending

Select-Object -First 10 Name, CPU, WorkingSet | Selects the first 10 processes in the output, and only shows the fields Name, CPU, and WorkingSet

Export-CSV top_processes.csv -NoTypeInfoInformation Exports the results into a CSV file named: top_processes.csv

CSV = Comma-Separated Values, -NoTypeInfoInformation prevents PowerShell from adding extra metadata/header information to the CSV

Write-Host "Process report generated." This is like a print statement in other languages, sends that text to the terminal

To run the script: **.\process_report.ps1**

Now we are making a script that will export: active TCP connections, listening ports, and local IP info.